LIVING TEXTBOOK · DESIGN REPORT

How a chapter becomes a living lesson

Three lesson flows, how we split a chapter into episodes, and our automation pipeline

In one minute

1. **Three lesson flows.** Every concept is taught with one of three simple patterns: **Story**, **Challenge** or **Wonder**. Each one hooks the class with a real Indian example, connects it to the concept, explains it from the NCERT textbook, then checks understanding.
2. **Chapters become episodes.** A chapter is split into short episodes of 1–2 concepts, each fitting one class period, like a web series: *Opener* → *Episodes* → *Finale*.
3. **One automated pipeline, one app.** We give the system a chapter PDF; it plans, scripts, voices, draws, animates and packages the lesson. The player app never changes. Only the chapter content does.

Who it is for

Students	NCERT Classes 5–10. Pilot subjects: Science, Social Science, Maths
Setting	The teacher projects the lesson to the whole class. The class votes by show of hands; the teacher taps the majority answer.
Context	Rooted in rural India (Karnataka, Maharashtra, Gujarat first): local crops, food, weather, markets, festivals
Language	English first. Narration-based, so other languages are added later by re-recording the voice only
Visual style	Inspired by the NCERT illustrations, made richer. Indian characters appear on screen but don't speak; a narrator tells the story.

Why we changed the old flow

Old flow (Predict → Observe → Explain → Check)	New approach
3 questions asked before students saw anything	Hook first: a story, problem or surprising sight, then one question
Foreign examples (willow tree, lab beaker)	Indian, rural examples the child has already seen
Not mapped to the NCERT chapter	Follows the NCERT concepts and activities one to one
20 personalised video paths for one student	Whole-class vote; at most 2–4 short replies per question
Built for one topic only	Works for any subject and any chapter

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Prepared 4 November 2026 · Reference chapters: Class 7 Science Ch 10 "Life Processes in Plants", Social Science Ch 2 "Understanding the Weather", Maths Ch 3 "A Peek Beyond the Point"

1 The three lesson flows

All three follow the same teaching idea: **create interest → connect to the concept → explain from the textbook → check**. What changes is *where the interest comes from*.

Flow 1: Story

Interest from a character

Story Real-life scene with an Indian child; ends on a question → **Vote** "What do you think?" Class votes → **Explain** Textbook concept, 60–90 s → **Reveal** Whose guess was right, and why → **Check** 1–2 quick MCQs

Best for: every subject; strongest in Social Science.

Example (Social): Zoya's father, a fisherman in Kutch, wants to sail. The barometer reads 996 mb. *Sail or stay?* → air pressure explained.

Flow 2: Challenge

Interest from a problem

Challenge A real job: "Help the shopkeeper / farmer..." → **Try** Class attempts it with what they know → **Stuck** "We can't decide without knowing ___" → **Learn** Concept arrives as the missing tool → **Solve** Back to the challenge, now solved → **New challenge** Same idea, new situation (MCQ)

Best for: Maths and practical Social topics. Being stuck first makes the explanation matter (research calls this "productive failure").

Example (Maths): Pooja keeps score on sports day. Priya jumped 3.5 m, Anu 3.45 m. *Who won?* → comparing decimals.

Flow 3: Wonder

Interest from a surprise

Wonder A surprising sight, 10–15 s, no narration → **Question** Class picks what they want to find out → **Investigate** The NCERT activity: live, simulated or on video → **Explain** Answers the class's question → **Apply** MCQ + "spot it near your home"

Best for: Science and other activity-heavy chapters. It matches the NCERT approach of asking questions first.

Example (Science): Grass under a lifted stone is yellow in a green field. *Why?* → test with 3 pots → plants need light to make food.

Comparing the three flows

	Story	Challenge	Wonder
Hook	A character's situation	A problem to solve	A surprising sight
Child's role	Listener who guesses	Problem-solver	Investigator
Activities sit	Mostly at the end	At the end	In the middle (core)
Best subject fit	All; strongest in Social	Maths, practical Social	Science
Teacher effort	Lowest	Medium	Highest
Video needed	Most	Medium	Least

> **Rule:** the wrong options in every vote are **common student misconceptions** (for example "plants eat soil" or "3.45 is bigger than 3.5"), never random wrong answers. The explanation then shows why each one is wrong.

1 The flows in practice: one concept from each subject

The same concept-loop applied to three real NCERT Class 7 chapters.

Social Science · Flow 1 Story · Concept: air pressure

Screen	What the class sees and hears	Media	Time
Story	A fishery office in Kutch. The barometer reads 996, the wind sock is stiff. "Abba wants to sail at dawn."	Video	25 s
Vote	Sail or stay? Class votes.	Still + timer	15 s
Explain	Air has weight; about 1013 mb is normal at the coast; below 1000 mb is a "depression" and a storm may form.	Video	60 s
Reveal	"Stay" is correct: "You read the signs like a meteorologist."	Still	8 s
Check	Which reading should worry a fisherman: 1015 / 1008 / 992?	MCQ	20 s

Maths · Flow 2 Challenge · Concept: comparing decimals

Screen	What the class sees and hears	Media	Time
Challenge	Sports day long jump final. Priya: 3.5 m. Anu: 3.45 m. Pooja must announce the winner.	Video	20 s
Try	Who won? Class votes.	Still + timer	15 s
Stuck	"Half of you said Anu, because 45 is bigger than 5. Who is right?"	Still + voice	10 s
Learn	Tape zooms in; place-value chart compares 3.50 and 3.45 place by place.	Video	50 s
Solve	Priya wins by 5 hundredths of a metre, about the width of your finger.	Video	10 s
New challenge	Pick the bigger one: 2.7 or 2.65 · 0.5 or 0.45 · 1.09 or 1.1	MCQ	40 s

Science · Flow 3 Wonder · Concept: leaves make starch

Screen	What the class sees and hears	Media	Time
Wonder	A hand turns over a croton leaf that is half green, half white. No narration.	Video	12 s
Question	"Do both parts make food?" · "Why is part white?" · "Where is food stored?"	Still	20 s
Investigate	Do-along: Betadine on potato, cooked rice and the leaf. The screen shows steps and a timer.	Activity screen	4 min
Explain	Blue-black = starch. Only the green parts make it; chlorophyll catches sunlight. Leaves are "food factories".	Video	60 s
Apply	A red amaranth (lal maath) leaf turns blue-black. What does it mean?	MCQ	20 s

Making it Indian and local

Instead of…	We use…
Willow tree, pond weed in a lab beaker	Chilli nursery, hydrilla in the village pond, sugarcane, moong sprouts
Generic city weather	Kutch fishermen, monsoon signs from elders (ants carrying eggs), IMD alerts
Abstract numbers	Sports day, kirana bills, market weights (0.2 kg capsicum)
Lab-only experiments	Safe home versions: Betadine iodine test, brick on grass, bottle rain gauge

2 Dividing a chapter into episodes

A chapter has many concepts, too many for one sitting. We split it like a **web series**: an opener that sets up the story world, short episodes that each fit one class period, and a finale.

Structure of a chapter

CHAPTER
e.g. Science Ch 10 "Life Processes in Plants" · 7 concepts

↓

OPENER (once, 30–60 s)
Introduces the story world or the big question of the chapter

↓

Episode 1	Episode 2	Episode 3	Episode 4
Concepts 1 + 2	Concepts 3 + 4	Concepts 5 + 6	Concept 7
Class period 1	Class period 2	Class period 3	Class period 4

↓

FINALE
Story payoff (the problem is solved) → practice activities → chapter MCQ round

Inside every episode

Recap (15 s)
"Last time..." + 1 recall question

CONCEPT LOOP · REPEATED FOR 1–2 CONCEPTS · USES THE CHOSEN FLOW

Hook Story / challenge / wonder → **Vote** Class decides → **Explain** Textbook concept → **Check** MCQ

Cliffhanger (10 s)
"But the plant in the jar is still weak. What's missing? Next time..."

The recap helps students remember: recalling last lesson's idea is one of the strongest ways to make learning stick. The cliffhanger gives the class a reason to come back.

Not every concept gets the same time

Concept size	Treatment	Time	Examples
Big / core	Full loop: hook → vote → explain → check	3–5 min	Photosynthesis, comparing decimals, air pressure
Small / supporting	Mini loop: one hook line → explain → one check	1–2 min	What weather is, units of length
Fact box	"Did you know?" card between loops, no vote	20 s	Know a Scientist, Vrikshayurveda, IMD motto

Keeping it fresh: vary what fills each step

Loop	Hook	Vote	Explain	Check
1	Story scene	2 cards	Narrated animation	MCQ
2	Silent "wonder" shot	3 cards	Live activity first	"What did you see?"
3	Challenge	Decision (sail / stay)	Diagram builds up	Odd one out
4	Surprising fact	True / false	Simulator	Match the pairs

2 Example: Science Ch 10 as episodes

Step 1: list the concepts from the chapter

#	Concept	NCERT section / activity	Size	Flow
1	Plants need light and water to grow	10.1 · Act 10.1 (three pots)	Big	Wonder
2	Leaves make starch; chlorophyll	10.2.1 · Act 10.2, 10.3 (iodine test)	Big	Wonder
3	Carbon dioxide from air is needed	10.2.2 · Act 10.4	Big	Challenge
4	Oxygen is released + word equation	10.2.2–10.2.3 · Act 10.5	Big	Wonder
5	Stomata	10.2.4 · Act 10.6	Small	Mini loop
6	Transport: xylem and phloem	10.3 · Act 10.7 (red ink)	Big	Story
7	Respiration in plants	10.4 · Act 10.8 (moong seeds)	Big	Wonder
–	R.H. Dastur, Kamala Sohonie, Vrikshayurveda	Fact boxes	Fact	"Did you know?"

Step 2: group the concepts into episodes

Period	Episode	What plays	Screen time
1	Opener + Ep 1	Opener → Light (full) → Starch (full, live Betadine test) → cliffhanger	~15 min + activity
2	Ep 2	Recap → CO_2 (full) → Oxygen + equation (full) → "Did you know: R.H. Dastur" → cliffhanger	~12 min
3	Ep 3	Recap → Stomata (mini) → Transport (full, red ink set up live) → cliffhanger	~10 min
4	Ep 4 + Finale	Recap → Respiration (full) → story payoff → activities → 10 MCQs	~15 min

> Each period uses **10–15 minutes of screen time**, leaving the rest of a 40-minute class for the teacher's discussion and activities.

Step 3: sort the activities

Each NCERT activity is assigned to one of three activity modes, based on time, safety and cost:

Do-along (live)
Quick, safe and cheap: the class does it now, with steps and a timer on screen
Betadine test, red ink in sadabahar, measuring with a ruler

Simulator (on screen)
Too slow, dangerous or costly for class: sliders and toggles
3 pots over 2 weeks, caustic-soda bottle, barometer dial

Village mission (home)
Takes days or needs family / community
Brick on grass, rain gauge for a month, ask elders for weather sayings

The chapter map on screen

A progress path is always visible; the teacher can jump to any concept.



The same method works for every subject

	Science Ch 10	Social Ch 2 (Weather)	Maths Ch 3 (Decimals)
Concepts	7	7	7
Main flow	Wonder	Story	Challenge

Story world	Ravi's chilli nursery, Karnataka	Zoya's fishing family, Kutch, Gujarat	Pooja, sports-day scorekeeper, Maharashtra
Class periods	4	3	3
Video clips (approx.)	~10	~16	~15

3 Our automation flow

From a chapter PDF to a finished classroom lesson. Steps marked **orange** need a human decision; everything else is automated.

1. **Intake**: upload the NCERT chapter PDF. — User
2. **Analyse**: find the subject, concepts, activities, in-text questions, fact boxes and end-of-chapter questions. Size each concept (big / small / fact). — Automated
3. **Plan**: split into episodes (1–2 concepts per period); choose a flow per concept (Story / Challenge / Wonder); choose the Indian setting and story world; sort activities into Do-along / Simulator / Mission. — Automated
4. ✓ **Checkpoint**: approve the episode plan before anything is produced. Changes here are cheap; changes after rendering are expensive. — Team review
5. **Script**: narration lines, vote options (built from misconceptions), MCQs, recap and cliffhanger lines. — Automated
6. **Narration audio**: choose the narrator voice → generate with ElevenLabs → measure each line's length (timing is built around the audio). — User picks voice
7. **Visuals**: style reference → characters, scenes and props → storyboards → still images (vote cards, MCQ images, fact cards). — Automated
8. **Animation** → **review** → **edit**: EN2 renders from storyboards, cut review, re-time or repair, stitch with narration. — Existing skill files
9. **Package**: write the chapter content file (`chapter.json`) + media folder → drop into the standard player → ZIP, ready to open offline. — Automated

Who does what

Step	Tool
Analyse, plan, script, package	Claude
Images, storyboards, stills	Codex image generation
Narration	ElevenLabs (through Codex)
Animation	EN2 video (Ultra)
Player app	One HTML player that works offline from a ZIP

3 One app for every chapter

The player app is built **once**. For each new chapter, in any subject, only the **content** changes. The app code is never edited.

PLAYER ENGINE (built once, never changes)

Knows how to show a fixed set of screen types:

Video · Vote · Reveal · MCQ · Explain · Do-along · Simulator · Mission · Recap · Cliffhanger · Did you know? · Chapter map · Finale / score

reads →

CHAPTER CONTENT (new for each chapter)

`chapter.json`: episodes → concepts → list of screens

`media/`: video clips, narration audio, still images

```
{ "concept": "Comparing decimals",
"flow": "challenge",
  "screens": [
    { "type": "video", "src":
"media/c6_challenge.mp4" },
    { "type": "vote",  "q": "Who won?",
      "options": ["Priya 3.5 m", "Anu
3.45 m"] },
    { "type": "video", "src":
"media/c6_learn.mp4" },
    { "type": "mcq",   "q": "Which is
bigger?",
      "options": ["2.7", "2.65"],
"answer": 0 } ] }
```

The three flows are just different orders of the same screens

Flow	Screen sequence
Story	Video → Vote → Explain → Reveal → MCQ
Challenge	Video → Vote → Stuck → Explain → Solve video → MCQ
Wonder	Silent video → Question cards → Do-along / Simulator → Vote → Explain → MCQ

Reusable simulators, set up from the content file

Simulators are the only interactive parts. To avoid writing new code per chapter, we build a small set of templates and configure them with data:

Template	Science	Social	Maths
Compare setups (A / B / C toggles)	3 pots, CO_2 bottle	Wet cloth open vs closed	–
Zoom scale / number line	–	Rain gauge	Ruler: tenths → hundredths
Dial / gauge	–	Barometer, thermometer, humidity	–
Tap the map / diagram	Plant parts	IMD warning map	Place-value chart

> If an activity doesn't fit any template, it is shown as a short video instead, so **no new code is needed per chapter**.

What this gives us

New chapter	New `chapter.json` + media; same app

New subject	Same three flows, same screen types
New language	Re-record narration and translate the text in `chapter.json`; videos are reused (no on-screen text, no lip-sync)
Offline classrooms	The whole lesson runs from a ZIP: open `index.html`, no internet needed

Next steps

#	Task
✓	Done: the general player engine with all screen types, the three activity modes and four simulators, and the `chapter.json` format (tested end to end)
✓	Done: skill files and pipeline scripts updated to the new pipeline (analyse → plan → checkpoint → script → … → package)
1	Produce one pilot episode: Science Ch 10, Episode 1 (Light + Starch), and test it in a real classroom
2	Pilot one Maths and one Social episode to confirm the method works across subjects